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Dr. Moufida Maimour, Dr. Samir Si-Mohammed, Pr. Hind Bril El Haouzi
CRAN Laboratory
Dr. Malcolm Egan
Inria Equipe Maracas

Subject: Application for PhD Position – Frugal Data Governance for Digital Twins (PC4)

Dear Dr. Maimour, Dr. Si-Mohammed, Pr. Bril El Haouzi, and Dr. Egan,

I am highly motivated to apply for the PhD position on Frugal and AI-Enhanced Data Governance for Reliable Digital Twins. As a computer science engineer specializing in Machine Learning and Edge-Optimized architectures, the challenge of shifting from volume-based data logic to information frugality in dynamic cyber-physical systems aligns perfectly with my research trajectory.

The challenge of feeding digital twins with reliable data under distributed, energy-limited constraints is an issue I confronted extensively during my engineering of the Industrial Gas Turbine Digital Twin. Recognizing that exhaustive telemetry was unsustainable, I designed an Edge-Optimized AI pipeline using Temporal Convolutional Networks (Edge-TCN) to intelligently sample, filter, and compress data directly on constrained edge hardware. This approach allowed for highly accurate anomaly detection using only strictly relevant data streams. I am eager to scale this methodology to address your goals in intelligent sensor selection and robust wireless transmission. Furthermore, through my work in Scientific Machine Learning and GNNs, I have developed robust frameworks to quantify uncertainties and propagate confidence indicators, ensuring that data frugality does not compromise operational reliability.

I am excited about the opportunity to validate this research on real-world industrial platforms and contribute to the Artemis framework. Please find attached my resume and academic transcripts. I look forward to discussing how my technical competencies align with the CRAN and Inria Maracas laboratories.

Sincerely,

Ismail Aissaoui